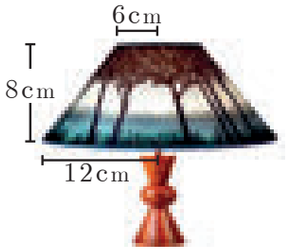
Exercise 7.1

1. The radius and height of a cylinder are in the ratio $5:7$ and its curved surface area is 5500 sq.cm. Find its radius and height.
2. A solid iron cylinder has total surface area of 1848 sq.m. Its curved surface area is five – sixth of its total surface area. Find the radius and height of the iron cylinder.
3. The external radius and the length of a hollow wooden log are 16 cm and 13 cm respectively. If its thickness is 4 cm then find its T.S.A.
4. A right angled triangle PQR where $\angle Q = 90^\circ$ is rotated about QR and PQ. If QR=16 cm and PR=20 cm, compare the curved surface areas of the right circular cones so formed by the triangle.
5. 4 persons live in a conical tent whose slant height is 19 m. If each person require 22 m^2 of the floor area, then find the height of the tent.
6. A girl wishes to prepare birthday caps in the form of right circular cones for her birthday party, using a sheet of paper whose area is 5720 cm^2, how many caps can be made with radius 5 cm and height 12 cm.
7. The ratio of the radii of two right circular cones of same height is $1:3$. Find the ratio of their curved surface area when the height of each cone is 3 times the radius of the smaller cone.
8. The radius of a sphere increases by 25%. Find the percentage increase in its surface area.
9. The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at ₹ 0.14 per cm^2.
10. The frustum shaped outer portion of the table lamp has to be painted including the top part. Find the total cost of painting the lamp if the cost of painting 1 sq.cm is ₹ 2.



7.3 Volume

Having discussed about the surface areas of cylinder, cone, sphere, hemisphere and frustum, we shall now discuss about the volumes of these solids.

Volume refers to the amount of space occupied by an object. The volume is measured in cubic units.

7.3.1 Volume of a solid right circular cylinder

The volume of a right circular cylinder of base radius 'r' and height 'h' is given by V = (Base Area) × (Height) = $\pi r^2 \times h = \pi r^2 h$ cubic units.

Therefore, Volume of a cylinder = $\pi r^2 h$ cu. units.

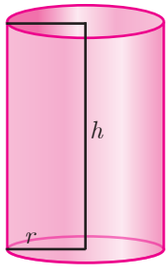


Fig. 7.25

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